



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Computer Science and Engineering

Academic Year 2025 – 2026 (Even Semester)

Degree, Semester & Branch: II Semester B.E. CSE 'B'

Course Code & Title: CS25C07 Object Oriented Programming

Name of the Faculty member (s): Dr.B.Vijayalakshmi, ASP-II/CSE

Innovative Practice Description

- Unit / Topic: Unit 1 / Principles of Object Oriented Programming

- Course Outcome: CO1

- Activity Chosen: Crossword Puzzle

- Justification:

Using a crossword puzzle as an active learning strategy for Object-Oriented Programming concepts encourages students to recall and connect important technical terms such as class, object, inheritance, polymorphism, encapsulation, and abstraction. The activity improves engagement, promotes peer learning, and helps students strengthen conceptual understanding in a fun and interactive manner.

- Time Allotted for the Activity: 10 Minutes

- Details of the Implementation:

- The basic OOPS concepts such as class, object, inheritance, polymorphism, abstraction, and encapsulation were explained to the students.
- After the completion of the lecture session, a crossword puzzle activity based on OOPS terminology was conducted.
- The puzzle sheet containing clues related to OOPS concepts was distributed to students in teams of two members to encourage collaborative learning.
- Students were instructed to complete the puzzle within 10 minutes.
- After completion, the answers were discussed and important concepts were reinforced through explanation and clarification of doubts.
- The activity improved active participation and conceptual understanding among students.
- The student's involvement in the activity is shown in Fig.1 and the sample answers given by the students is shown in Fig.2.

- CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO5	3	2	2	-	1	-	-	-	-	-	1	3	3	1

(1 – Low 2 – Moderate 3 – High)

- PO / PSO mapped:

Innovative practice	PO1	PO2	PO9	PO10	PSO1
	3	2	1	1	3
Justification for correlation	The students gain the knowledge of core object-oriented programming concepts..	Students analyze clues and identify appropriate OOPS terminology.	Students work collaboratively in teams to solve the puzzle.	Students communicate and discuss solutions effectively.	The activity strengthens programming and problem-solving skills related to software development.

- Images / Screenshot of the practice:

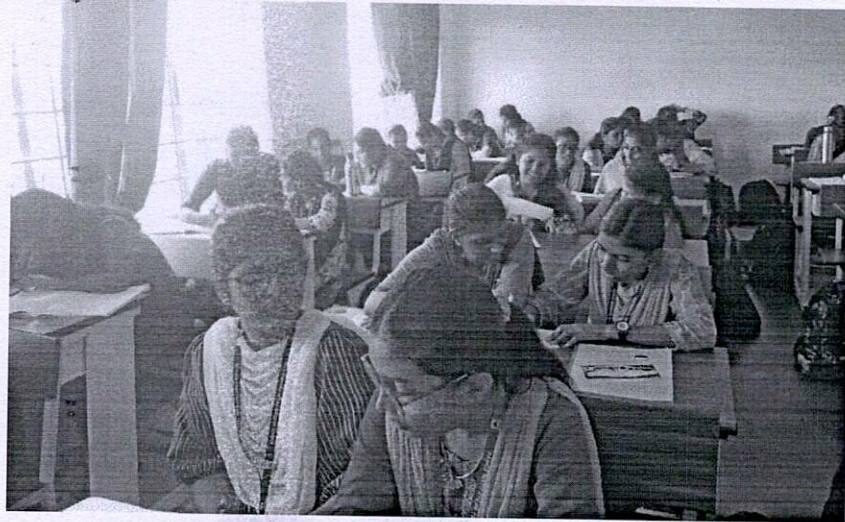
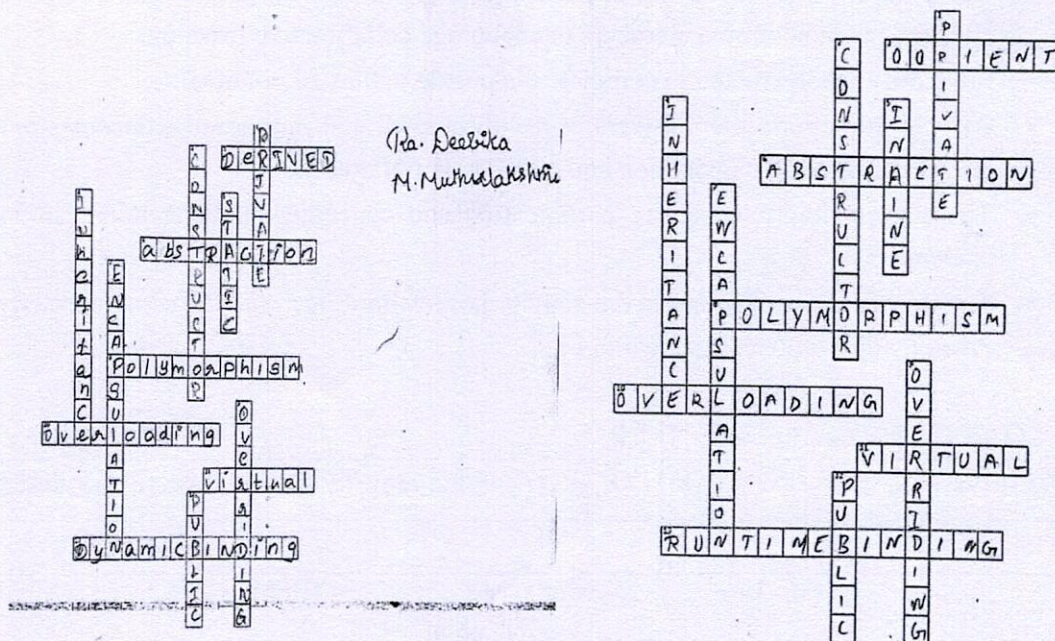


Fig 1. Student Involvement



S.S. Santhosh
S.Arun kumar

OOPS Concepts- Cross Word Puzzle

ACROSS

- 3. CLASS, Child class concept
- 6. Hiding internal implementation
- 8. Feature supporting multiple forms
- 10. Same function name with different parameters
- 11. Keyword enabling runtime polymorphism (C++)
- 13. Execution-time method resolution

DOWN

- 1. Access specifier restricting access within class
- 2. Special method used to initialize objects
- 4. Mechanism of acquiring properties from parent class
- 5. Keyword for class-level members
- 7. Wrapping data and methods together
- 9. Ability to redefine base class method
- 12. Access specifier allowing full access

Fig 2. Solved Crossword Puzzle

- **Reflective Critique:**

- ❖ *Feedback of practice from students and other stakeholders:*

- Students reported that this activity made learning OOPS concepts more interesting and interactive. The gamified approach increased motivation and participation during the session.

- ❖ *Benefit of the practice:* (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

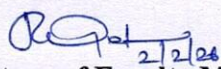
- The activity improved concept retention and helped students easily remember technical terminology.
- It also supported collaborative learning and active participation compared to traditional lecture-based teaching

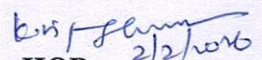
- ❖ *Challenges faced in implementation:*

- Some students initially found it difficult to identify the answers from the clues due to unfamiliar terminology. Additional hints and guidance were provided to support them.

References:

- <https://puzzlemaker.discoveryeducation.com/criss-cross>
- https://www.ritrjpm.ac.in/images/computer-science/2022-2023/2_GS_CS8651_Crossword_Puzzles.pdf


Signature of Faculty Member


HOD